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Lecture 31 9/8/26

Degrees of freedom: The # of coordinates that can be independently varied in small displacements

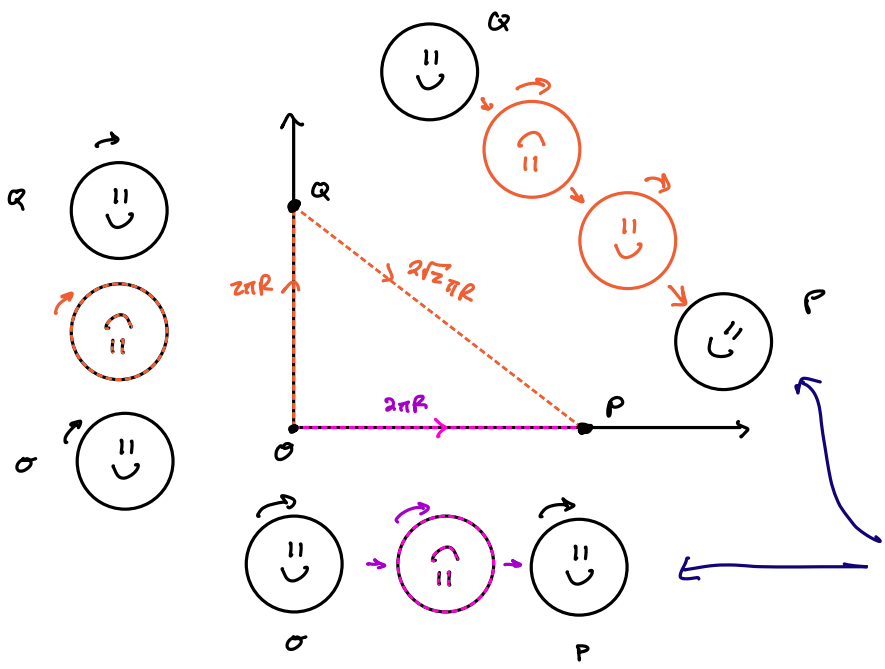
$\Rightarrow$ # of directions a system can move to change its configuration.

When the # of degrees of freedom $< 3N$, the system is <u>constrained</u>

When the # of degrees of freedom = the # of generalized coordinates, the system is <u>holonomic</u>

In this class, we focus on <u>holonomic</u> systems

Counter example: A ball rolling in 2D



These two states are not the same

$\Rightarrow$ we need > 2 coordinates even with only 2 deg. of freedom

You won't see non-holonomic systems like this in this class.

For holonomic systems w/ generalized coordinates $q_1, \dots, q_n$, the system's time evolution can be fully described by:

$$\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \qquad i = 1, \dots, n$$

Proof: Consider a single particle with two types of forces acting on it:

$$\vec{F}_{total} = \underbrace{\vec{F}_{constraint}}_{\text{Constraint forces: tension, Normal force, Static friction}} + \underbrace{\vec{F}}_{\text{other forces, like gravity, Lorentz, etc.}}$$

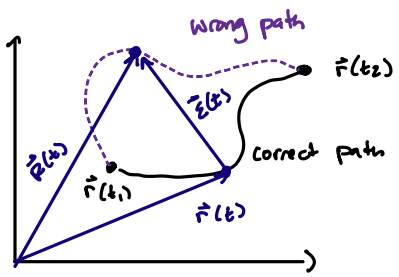
We must assume all $\vec{F}$ are conservative to proceed ($\vec{F}_{constraint}$ do not have to be conservative) (This means to kinetic friction, no drag, for example

$$\Rightarrow \vec{F} = -\vec{\nabla} U(\vec{r}, t)$$)

so we can write $\mathcal{L} = T - U$

We want to make the action stationary:

$$S = \int_{t_1}^{t_2} \mathcal{L}\, dt$$



$$\vec{R}(t) = \vec{r}(t) + \vec{\varepsilon}(t) \qquad \vec{\varepsilon} = 0 \quad \text{at } t = t_1, t_2$$

$$S = \int_{t_1}^{t_2} \mathcal{L}(\vec{R}, \dot{\vec{R}}, t)\, dt \qquad S_0 = \int_{t_1}^{t_2} \mathcal{L}(\vec{r}, \dot{\vec{r}}, t)\, dt \Rightarrow \text{the action of the stationary path}$$

Want to show: S is stationary when $\vec{R}(t) = \vec{r}(t)$ (or when $\vec{\varepsilon}(t) = 0$)

or, $\delta S = S - S_0 = 0$ when $\vec{R}(t) = \vec{r}(t)$

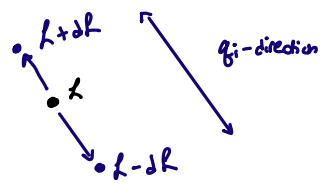
$$\delta S = S - S_0 = \int_{t_1}^{t_2} \mathcal{L}(\vec{R}, \dot{\vec{R}}, t)\, dt - \int_{t_1}^{t_2} \mathcal{L}(\vec{r}, \dot{\vec{r}}, t) = \int_{t_1}^{t_2} \underbrace{[\mathcal{L}(\vec{R}, \dot{\vec{R}}, t) - \mathcal{L}(\vec{r}, \dot{\vec{r}}, t)]}_{\equiv \delta \mathcal{L}}\, dt \equiv \int_{t_1}^{t_2} \delta \mathcal{L}\, dt$$

Quick note about notation:

$\mathcal{L}(\vec{r})$

$\delta \mathcal{L}$ comes from varying $\vec{r}$ in any direction (of constrained coordinates)

different than $\frac{\partial \mathcal{L}}{\partial q_i}$



$$\mathcal{L}(\vec{r}, \dot{\vec{r}}, t) = T - U = \tfrac{1}{2} m \dot{\vec{r}}^2 - U(\vec{r}, t)$$

$$\mathcal{L}(\vec{R}, \dot{\vec{R}}, t) = T - U = \tfrac{1}{2} m \dot{\vec{R}}^2 - U(\vec{R}, t) = \tfrac{1}{2} m (\dot{\vec{r}} + \dot{\vec{\varepsilon}})^2 - U(\vec{r} + \varepsilon, t)$$

$$= \tfrac{1}{2}m\dot{r}^2 + m\dot{\vec{r}}\cdot\dot{\vec{\varepsilon}} + \tfrac{1}{2}m\dot{\varepsilon}^2 - U(\vec{r}+\varepsilon, t) \approx \tfrac{1}{2}m\dot{r}^2 + m\dot{\vec{r}}\cdot\dot{\vec{\varepsilon}} - U(\vec{r}+\varepsilon,t) + O(\varepsilon^2)$$

$$\delta L = L(\vec{R}, \dot{\vec{R}}, t) - L(\vec{r}, \dot{\vec{r}}, t) = \cancel{\tfrac{1}{2}m\dot{r}^2} + m\dot{\vec{r}}\cdot\dot{\vec{\varepsilon}} - \cancel{\tfrac{1}{2}m\dot{\vec{r}}^2} - [U(\vec{r}+\varepsilon, t) - U(\vec{r}, t)]$$

<u>simplify $U(\vec{r}+\vec{\varepsilon}, t)$</u>: Taylor series for a function of a vector:

$$f(\overset{\vec{r}+\vec{\varepsilon}}{\vec{x}}) = f(\overset{\vec{r}}{\vec{a}}) + (\vec{x}-\vec{a})\cdot\vec{\nabla} f(\vec{a}) + O(\vec{x}-\vec{a})^2$$

$$U(\vec{r}+\vec{\varepsilon}) = U(\vec{r}) + \vec{\varepsilon}\cdot\vec{\nabla}U(\vec{r}) + O(\varepsilon^2)$$

$$\Rightarrow \delta L = m\dot{\vec{r}}\cdot\dot{\vec{\varepsilon}} - [U(\vec{r}) + \vec{\varepsilon}\cdot\vec{\nabla}U(\vec{r}) - U(\vec{r})] + O(\varepsilon^2)$$

$$\delta L = m\dot{\vec{r}}\cdot\dot{\vec{\varepsilon}} - \vec{\varepsilon}\cdot\vec{\nabla}U + O(\varepsilon^2) = m\dot{\vec{r}}\cdot\dot{\vec{\varepsilon}} - \vec{\varepsilon}\cdot\vec{\nabla}U$$

$$\Rightarrow \delta S = \int_{t_1}^{t_2} \delta L\, dt = \int_{t_1}^{t_2} [\underbrace{m\dot{\vec{r}}}_{u}\cdot\underbrace{\dot{\vec{\varepsilon}}}_{dv} - \vec{\varepsilon}\cdot\vec{\nabla}U]\, dt = \int_{t_1}^{t_2} [-\underbrace{m\ddot{\vec{r}}}_{du}\cdot\underbrace{\vec{\varepsilon}}_{v} - \vec{\varepsilon}\cdot\vec{\nabla}U]\, dt$$

$$= -\int_{t_1}^{t_2} \dot{\vec{\varepsilon}}\cdot[m\ddot{\vec{r}} + \vec{\nabla}U]\, dt$$

$$m\ddot{\vec{r}} = \vec{F}_{total} \Rightarrow \delta S = -\int_{t_1}^{t_2} \vec{\varepsilon}\cdot[\vec{F}_{total} + \nabla\vec{U}]\, dt = -\int_{t_1}^{t_2} \vec{\varepsilon}\cdot[\vec{F}_{constraint} + \underbrace{\vec{F} + \vec{\nabla}\vec{U}}]\, dt$$

cancel: $\vec{F} = -\vec{\nabla}U$

$$\delta S = -\int_{t_1}^{t_2} \vec{\varepsilon}\cdot\vec{F}_{constraint}\, dt = 0$$

since all constraint forces are $\perp$ to the surface of movement

($\vec{\varepsilon} \perp$ to all $\hat{q}_i$'s) $\quad \vec{\varepsilon} \perp \vec{F}_{constraint}$

$\Rightarrow \vec{\varepsilon}\cdot\vec{F}_{constraint} = 0$

$$\Rightarrow \delta S = 0 \quad ☑$$

$\Rightarrow$ We showed $\delta S = S - S_0 = 0$ up to $O(\varepsilon^2)$ when variations occur in the space of minimal generalized coordinates.

$S = \int_{t_1}^{t_2} L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n)\, dt$ being stationary implies:

$$\frac{\partial L}{\partial q_i} = \frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} \qquad i = 1, \dots, n \qquad \text{(by arguments in Ch. 6)}$$

$\Rightarrow$ For any holonomic system, with all non-constraint forces derivable from a potential energy U:

$$\frac{\partial L}{\partial q_i} = \frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i}$$ describes the motion of the system